

ESTIMATING PERIOD FROM SPARSE, NOISY TIMING DATA

Barry G. Quinn*, I. Vaughan L. Clarkson[†] and Robby G. McKilliam[‡]

* Department of Statistics, Faculty of Science, Macquarie University, Sydney, NSW, 2109, AUSTRALIA

barry.quinn@mq.edu.au

[†] School of Information Technology & Electrical Engineering, The University of Queensland, Qld., 4072, AUSTRALIA

v.clarkson@uq.edu.au

[‡] Institute For Telecommunications Research, University of South Australia, Mawson Lakes, SA, 5095, AUSTRALIA

robby.mckilliam@unisa.edu.au

ABSTRACT

The problem discussed in this paper is that of estimating the period of a sequence of periodic events when the measurements of the occurrence times are noisy and sparse. The problem is common to many signal processing applications, such as baud estimation from zero-crossings in telecommunications and in pulse repetition interval estimation in electronic support measures. Previous algorithms have been based on periodogram maximisation [1, 2], Euclidean algorithms [3–5], least-squares line search [6], lattice line search [7] and Gaussian maximum likelihood [8]. Until now, very little has been known about the asymptotic statistical properties of any such algorithm. In this paper, a new algorithm is proposed, based on a modified least-squares approach. Under very general properties, the estimators of the system parameters are shown to have excellent (theoretical) asymptotic statistical properties. These properties are illustrated using a number of simulations.

Index Terms— Period estimation, pulse repetition interval, nearest lattice point problem, modified least squares.

1. INTRODUCTION

In this paper, we consider a signal model in which there is an underlying period to observed events. The times at which the events occur are measured with random errors and the events do not occur, or are not observed, at every consecutive period. That is, the record of measured occurrence times, or *times of arrival (TOAs)*, is sparse and noisy. From the TOAs, we seek to estimate the underlying period and its phase with respect to some time origin. This problem is common to baud estimation from zero crossings or frequency hops in telecommunications, pulse repetition interval estimation in electronic support measures and even beat tracking in music signal processing.

Fogel & Gavish [1] proposed a mathematical model which will form the basis of our investigations. The TOA of the n th event, X_n , is assumed to satisfy

$$X_n = \alpha + \beta S_n + \varepsilon_n; \quad n = 1, \dots, N, \quad (1)$$

where α is the phase, β is the period and the ε_n assumed to be Gaussian white noise. If the consecutive events are labelled by the integers then the S_n indicate those events that are represented in the

TOA record. We are primarily interested in estimating the parameters α and β , the S_n being treated as nuisance parameters.

Fogel & Gavish observe that $e^{j2\pi f X_n}$ has approximately the same phase for all n if $f \approx 1/\beta$. Therefore the sum over n of such exponentials should add constructively. They propose maximising the magnitude of the sum with respect to f in order to estimate β . The magnitude of the sum, as a function of f , is analogous to the periodogram for stochastic processes [9].

Casey & Sadler [4, 5] make the observation that the time differences $X_{n_1} - X_{n_2}$ cancel the phase and so are approximately commensurate to the period β . Under certain conditions, the period is in some sense the approximate greatest common measure of the time differences. Casey & Sadler adapt a multi-dimensional Euclidean algorithm to estimate the period by successive ordered differencing.

Clarkson *et al.* [3] assumes the increments $S_n - S_{n-1}$ are independent and geometrically distributed. Like Casey & Sadler, they develop a multi-dimensional Euclidean algorithm to estimate β .

Sidiropoulos *et al.* [6] develop a number of algorithms from consideration of least-squares estimation of the parameters. It turns out that these algorithms can be interpreted geometrically as maximisation of a certain function constrained to a line segment and for this reason they are termed ‘line search’ algorithms. They also consider the problem of identifiability. When the S_n are known, the problem reverts to simple linear regression. This gives rise to what they define as the *clairvoyant* Cramér-Rao lower bound (CRLB), the lower bound for unbiased estimation of period or phase when the S_n are assumed known.

Clarkson [7] shows that when *a priori* bounds are placed on α and β in (1) to ensure identifiability, Gaussian ML estimators exist. McKilliam & Clarkson [8] develop a practical, guaranteed Gaussian ML estimation algorithm.

Although, through the clairvoyant CRLB, the statistical behaviour of estimators has been studied in low-noise conditions, there has been no study of the asymptotic properties, that is, as the number of TOAs becomes large. In this paper, we drop most of the distributional assumptions and propose a method based on least squares. The estimator may be shown to be strongly consistent and satisfy a central limit theorem. Some of the methods of proof are similar to those introduced by McKilliam *et al.* [10].

The paper is organised in the following way. In Section 2, we set out further details of our statistical model. In Section 3, we propose a new modified least-squares estimator which we demonstrate through numerical simulations in Section 4. It is this estimator that, in Section 5, we show is strongly consistent and obeys a central limit theorem.

Vaughan Clarkson is on study leave at the Institute for Telecommunications, Vienna University of Technology, Austria

2. THE MODEL

In this paper we shall assume that $\{X_n\}$ satisfies (1), subject to the following additional constraints.

1. The period β is known to lie within one octave so that $\beta_L < \beta \leq 2\beta_L$ for some known $\beta_L > 0$ and $\alpha \in (-\beta/2, \beta/2)$.
2. Let $u_n = \varepsilon_n/\beta_0$, where β_0 is the true value of β . Then the u_n are continuous, independent and identically symmetrically distributed with mean 0 and finite absolute moment of order $2 + \nu$, where $\nu > 0$, and the $\langle u_n \rangle$ have common variance σ^2 and common probability density function f , unimodal with mode at 0. We adopt the notation $\langle x \rangle = x - \lfloor x + 1/2 \rfloor$, the difference between x and its closest integer.
3. $S_n = S_1 + \sum_{j=2}^n Y_j$, where S_1 is an unknown integer, and the Y_j are independent and identically distributed (i.i.d.) and discrete, with mean μ_Y , variance σ_Y^2 , finite sixth moment and common characteristic function $\phi_Y(t) = Ee^{itY}$ having the property that $\phi_Y(t) = 1$ only if t is a multiple of 2π .

$\{\varepsilon_n\}$ is *not* assumed to be Gaussian, nor $\{Y_n\}$ geometric.

A discussion of the above assumptions is in order.

- If β is unconstrained, we could replace β and S_n with β/k and kS_n for any integer k , and (1) would not change, since $\{kS_n\}$ would still be an integer-valued sequence.
- For fixed β , if the S_n were unconstrained integers, S_n could be replaced by $S_n + k$ for each n , and α by $\alpha - \beta k$, and (1) would again remain unchanged. The arbitrariness of S_1 explains why we are able to constrain α as we have done in Condition 1. See Remark 9 for further discussion.
- The unimodality condition ensures identifiability.
- The third condition imposes minimal distributional assumptions on the times between observations. The condition on $\phi_Y(t)$ precludes, for example, the possibility that the Y_j take on only even values.

3. ESTIMATION METHOD

If the S_n were known, α and β could be estimated by least squares, *i.e.*, by minimising

$$SS(\alpha, \beta) = \sum_{n=1}^N (X_n - \alpha - \beta S_n)^2.$$

When the S_n are unknown and must be estimated, this approach yields poor estimators. An example is shown in the top half of Figure 1, where the function $\min_{\alpha, S_1, \dots, S_N} SS(\alpha, \beta)$ is shown for a simulated sample. Although there is a local minimum near $\beta = 1$, the function appears to be generally increasing. In fact, the function is well-fitted by a quadratic, which is explained theoretically in Lemma 8. To obtain good performance, we instead minimise

$$SSMOD(\alpha, \beta) = \min_{S_1, \dots, S_N} \beta^{-2} \sum_{n=1}^N (X_n - \alpha - \beta S_n)^2. \quad (2)$$

A reparametrization will simplify the analysis. Put

$$\gamma = 1/\beta \in \left[\frac{1}{2\beta_L}, \frac{1}{\beta_L} \right), \quad \nu = \alpha/\beta,$$

where ν can lie in any prescribed interval of unit length, and consider the function

$$\sum_{n=1}^N (\gamma X_n - \nu - S_n)^2,$$

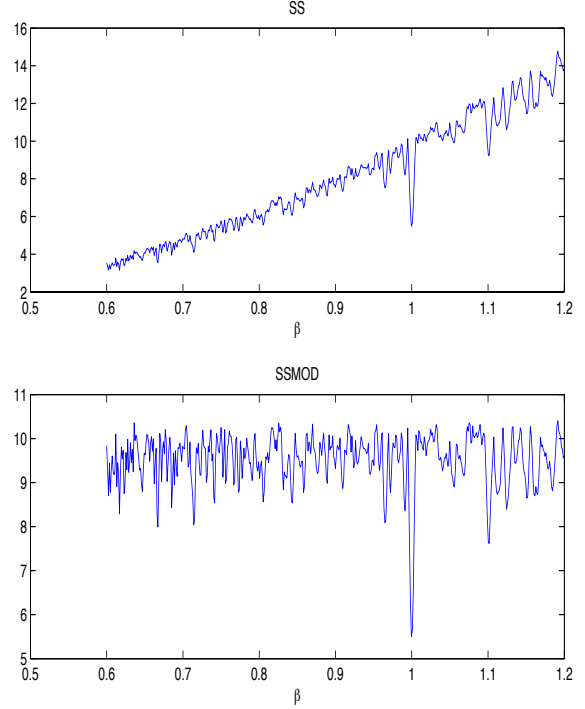


Fig. 1. Raw and modified sum of squares function with $N = 128$.

Now, for fixed γ and ν , this is minimised when, for each n , S_n is the closest integer to $(\gamma X_n - \nu)$. We thus only need to minimise $\sum_{n=1}^N \langle \gamma X_n - \nu \rangle^2$. Note that we are not here imposing the constraint $S_n \geq S_{n-1}$, but merely trying to find an objective function whose minimiser will provide an estimator of β .

Let $u_n = \varepsilon_n/\beta_0$, $\delta = \gamma\beta_0 - 1$ and $\zeta = \langle \gamma\alpha_0 - \nu \rangle$, where (α_0, β_0) denotes the true value of (α, β) . Let

$$V_N(\delta, \zeta) = \frac{1}{N} \sum_{n=1}^N \langle \gamma X_n - \nu \rangle^2 = \frac{1}{N} \sum_{n=1}^N \langle \zeta + \delta S_n + (1 + \delta)u_n \rangle^2$$

and let

$$(\hat{\delta}_N, \hat{\zeta}_N) = \arg \min_{\substack{\delta \in [-1/2, 1] \\ \zeta \in [-1/2, 1/2]}} V_N(\delta, \zeta).$$

We shall show later that $(\hat{\delta}_N, \hat{\zeta}_N)$ converges almost surely to $(0, 0)$, and derive the rate of convergence. This will establish the asymptotic results for the minimiser $(\hat{\alpha}_N, \hat{\beta}_N)$ of (2).

4. SIMULATIONS

The following figures summarise the results of simulations for various combinations of N and $\sigma_u = \sqrt{\text{var } u_n}$ and for two different Y_j distributions. In each case, $\beta_L = 0.7$, $\beta = 1.0$, the u_n were (simulated) Gaussian with mean 0 and variance $\sigma_u^2 = (0.05)^2 \times 2^{k/3}$, $k = -4, \dots, 18$, and the Y_j were

- geometric (starting at 0) with mean 1,
- geometric (starting at 0) with mean 3.

The estimation technique used was the exact method proposed in [8], modified to minimise (2), labelled ‘Modified LLS’. For each

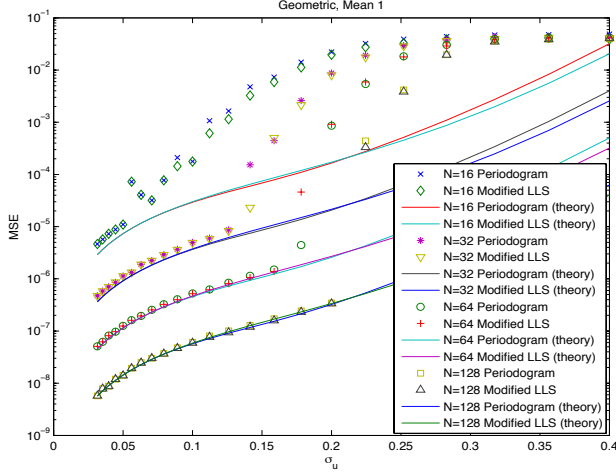


Fig. 2. Simulation results: Geometric, mean 1

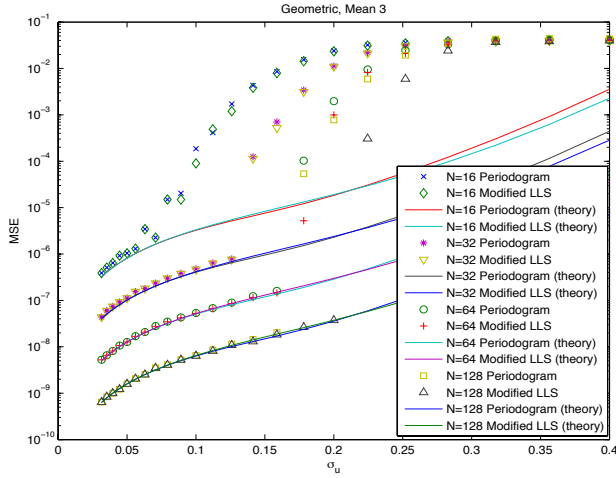


Fig. 3. Simulation results: Geometric, mean 3

combination of N and σ_u , 2000 replications were undertaken, and the sample mean square errors of the $\hat{\beta}_s$ computed. It is clear from the plots that the theory and simulations agree very well as σ_u increases, until a thresholding effect is noticeable, occurring as expected at values of σ_u , which depend on N .

In each case, a comparison is made with the periodogram method of Fogel & Gavish [1, 2]. The theoretical results which are presented are the subject of an independent, companion paper [11]. For the simulation results, we see close agreement between the modified LLS and periodogram estimators.

Although not presented here for want of space, results for simulations using the Poisson instead of the geometric distribution are similar, and also agreed with the theory.

5. STATISTICAL PROPERTIES

The following two theorems and corollary describe the main results of this paper. Their proofs are technical, and are omitted or briefly sketched due to space restrictions.

Theorem 1 $(N\hat{\delta}_N, \hat{\zeta}_N) \rightarrow (0, 0)$, almost surely as $N \rightarrow \infty$.

Corollary 2 $\hat{\alpha}_N - \alpha_0$ and $N(\hat{\beta}_N - \beta_0)$ converge almost surely to 0 as $N \rightarrow \infty$.

Theorem 3 The distribution of

$$\begin{bmatrix} N^{1/2}(\hat{\alpha}_N - \alpha_0) \\ N^{3/2}(\hat{\beta}_N - \beta_0) \end{bmatrix}$$

converges as $N \rightarrow \infty$ to the normal with mean 0 and covariance matrix

$$\frac{12\sigma^2\beta_0^2}{(1-h)^2\mu_Y^2} \begin{bmatrix} \frac{1}{3}\mu_Y^2 & -\frac{1}{2}\mu_Y \\ -\frac{1}{2}\mu_Y & 1 \end{bmatrix}, \quad (3)$$

where $h = f(-1/2)$.

Note: as the Gaussian “clairvoyant” Cramér-Rao lower bound [6] for unbiased estimators of β , that is, the Cramér-Rao lower bound when $\{\varepsilon_t\}$ is Gaussian and the $\{S_n\}$ are known constants, is

$$\frac{\beta_0^2 \text{var } u_n}{N \sum_{n=1}^N S_n^2 - \left(\sum_{n=1}^N S_n \right)^2},$$

the ratio of the asymptotic variance of our estimator to the Gaussian “clairvoyant” Cramér-Rao lower bound may be shown to converge almost surely to

$$(1-h)^{-2} \frac{\text{var } \langle u_n \rangle}{\text{var } u_n}$$

under our assumptions on the $\{S_n\}$.

It is usually the case that an estimator may be expressed as the minimiser of some function of the parameter and the data, which converges in some sense uniformly in the parameter on some compact set as $N \rightarrow \infty$ to a function of the parameters which is continuous, differentiable and uniquely minimised at the true parameter. This is far from the case here, where the limiting function is not continuous anywhere, as shown in Lemma 8. We have developed novel techniques to establish the asymptotic properties. The first key result is given in Lemma 6, which is established in two parts by using the results in Lemmas 4 and 5. We show that convergence of $V_N(\delta, \zeta)$ to its expectation occurs at all points on a fine grid, and then show that the maximum deviation between points off the grid and nearby grid points is sufficiently small to ensure the convergence at all points.

Lemma 4 Let

$$Q_N(\delta, \zeta) = EV_N(\delta, \zeta) | Y_2, \dots, Y_N.$$

For some $d > 0$, let

$$\delta_j = -\frac{1}{2} + \frac{3j}{2 \lfloor N^{d+1} \rfloor}, \quad j = 0, 1, \dots, \lfloor N^{d+1} \rfloor - 1$$

$$\zeta_k = -\frac{1}{2} + \frac{k}{\lfloor N^d \rfloor}, \quad k = 0, 1, \dots, \lfloor N^d \rfloor - 1$$

Then

$$\sup_{j,k} |V_N(\delta_j, \zeta_k) - Q_N(\delta_j, \zeta_k)| \rightarrow 0,$$

almost surely as $N \rightarrow \infty$.

Lemma 5 Let

$$T_N(\delta, \zeta) = EV_N(\delta, \zeta).$$

Then, with δ_j and ζ_k defined in Lemma 4,

$$\sup_{j,k} |Q_N(\delta_j, \zeta_k) - T_N(\delta_j, \zeta_k)| \rightarrow 0,$$

almost surely as $N \rightarrow \infty$.

Lemma 6 Let $W_N(\delta, \zeta) = V_N(\delta, \zeta) - T_N(\delta, \zeta)$. Then

$$\sup_{\delta, \zeta} |W_N(\delta, \zeta)| \rightarrow 0,$$

almost surely as $N \rightarrow \infty$.

Lemma 7 Let $\phi_u(t) = Ee^{itu_1}$. Then

$$T_N(\delta, \zeta) = c_0 + \sum_{|k|=1}^{\infty} c_k e^{i2\pi k \zeta'} \phi_u(2\pi k(1+\delta)) \psi_N(2\pi k(1+\delta)) \\ + O\left(\frac{1}{N}\right),$$

where $\zeta' = \zeta + \delta S_1$ and

$$\psi_N(x) = \begin{cases} \frac{\phi_Y(x) \{ \phi_Y(x) \}^{N-1} - 1}{N \{ \phi_Y(x) - 1 \}} & ; \quad \langle x \rangle \neq 0 \\ 1 & ; \quad \langle x \rangle = 0. \end{cases}$$

Lemma 8 For fixed (δ, ζ) ,

$$T_N(\delta, \zeta) \rightarrow T(\delta, \zeta),$$

where

$$T(\delta, \zeta) = \begin{cases} \frac{1}{12} & ; \quad \delta \text{ irrational} \\ E \langle u_1 + \zeta \rangle^2 & ; \quad \delta = 0, \end{cases}$$

while, if $\delta = p/q$, where $(p, q) = 1$ and $p \neq 0$,

$$T(\delta, \zeta) = \frac{1}{12} + \frac{1}{q^2} \left\{ E \langle (p+q)u_1 + q\zeta \rangle^2 - \frac{1}{12} \right\},$$

if q is odd and

$$\frac{1}{12} + \frac{1}{q^2} \left\{ E \left\langle (p+q)u_1 + q\zeta - \frac{1}{2} \right\rangle^2 - \frac{1}{12} \right\},$$

if q is even.

Proof. (Theorem 1) It may be shown using the above lemmas and complicated probabilistic argument that $(0, 0)$ is the only limit point of the sequence $\{(\hat{\delta}_N, \hat{\zeta}_N)\}$, for almost all points in the sample space. Once this is done, the almost sure convergence of $N\hat{\delta}_N$ to 0 may be established. The details are available from the authors. ■

Corollary 2 follows immediately.

Proof. (Theorem 3) The proof proceeds much in the same way as that of Theorem 1 in [10]. We put $\tilde{S}_N = S_N + 1$, so as to avoid the improbable but possible event that $S_N = 0$. After showing that

$$0 = \frac{1}{N} \sum_{n=1}^N \left[\frac{1}{\tilde{S}_N^{-1} S_n} \right] \left\langle \hat{\zeta}_N + \hat{\psi}_N \tilde{S}_N^{-1} S_n + \left(1 + \tilde{S}_N^{-1} \hat{\psi}_N \right) u_n \right\rangle,$$

and using the fact that

$$n^{-1} S_n = \frac{1}{n} S_0 + \frac{1}{n} \sum_{j=2}^n (Y_j - \mu_Y) + \frac{n-1}{n} \mu_Y,$$

to show that

$$N^{-1} S_N = \mu_Y + O_P(N^{-1/2}) \\ N^{-2} \sum_{n=1}^N S_n = \frac{1}{2} \mu_Y + O_P(N^{-1/2}) \\ N^{-3} \sum_{n=1}^N S_n^2 = \frac{1}{3} \mu_Y^2 + O_P(N^{-1/2}),$$

we establish the result that $N^{1/2} \begin{bmatrix} \hat{\zeta}_N & \tilde{S}_N \hat{\delta}_N \end{bmatrix}'$ is asymptotically normal with mean 0 and covariance matrix

$$\frac{12\sigma^2}{(1-h)^2} \begin{bmatrix} \frac{1}{3} & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{bmatrix}.$$

It is then shown that $\begin{bmatrix} N^{1/2}(\hat{\alpha}_N - \alpha_0) & N^{3/2}(\hat{\beta}_N - \beta_0) \end{bmatrix}'$ is asymptotically normal with mean 0 and covariance matrix as given in (3). ■

Remark 9 Our model allows for the true value of phase, α_0 , to be contained in an open interval of length β . We can close the interval at one end, for instance, to include $\alpha_0 = -\beta_0/2$, and obtain essentially the same theoretical results if we modify Corollary 2 and Theorem 3 slightly. We must replace $\hat{\alpha}_N - \alpha_0$ in their statements with $\beta_0 \langle (\hat{\alpha}_N - \alpha_0) / \beta_0 \rangle$.

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